12 Expand and simplify $(x - 2)(3x + 2)(2x + 3)$

$(x - 2)(3x + 2) = 3x^2 - 4x - 4$

$(3x^2 - 4x - 4)(2x + 3)$

$= 6x^3 + 9x^2 - 8x^2 - 12x - 8x - 12$

$= 6x^3 + x^2 - 20x - 12$

1 mark
expand first two brackets

1 mark
multiply third bracket

1 mark
simplify correctly

$6x^3 + x^2 - 20x - 12$

(Total for Question 12 is 3 marks)

13 In a school there are 16 teachers and 220 students.
Of these students 120 are girls and 100 are boys.

One teacher, one girl and one boy are going to be chosen to represent the school.

Work out the number of different ways there are to choose one teacher, one girl and one boy.

(Total for Question 13 is 2 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA